Permitless Concealed Carry Adoption and Firearm Mortality in U.S. States, 1999–2024: A Source-Audited Panel Analysis

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Abstract

Permitless concealed carry laws remove the requirement that eligible adults obtain a carry permit before carrying a concealed handgun in public. These laws spread rapidly during the last decade, but the empirical question is difficult because recent adopters are politically and structurally patterned, legal coding contains edge cases, and permitless carry often changes a bundle of pre-carry screens rather than a single mechanism. We assembled a 50-state annual panel from 1999 through 2024 using CDC mortality outcomes and a source-audited permitless-carry treatment file that records clean adopters, non-adopters, baseline-permitless states, partial cases, and ambiguous cases. Firearm suicide was the prespecified primary outcome. In the baseline state and year fixed-effects model, adjusted for unemployment and per-capita income with state-clustered standard errors, permitless carry adoption was associated with a 1.391 deaths per 100,000 increase in firearm-suicide rates. Age-adjusted firearm-suicide models gave a similar estimate (1.407). The association remained positive in fractional-year treatment coding, stacked DiD, covariate-balanced TWFE, population-weighted models, COVID-period exclusions, external firearm-law covariate models, leave-one-adopter-out checks, legal edge-case sensitivity analyses, and cohort-window ATT estimates, although the staggered-adoption and covariate-balanced estimates were smaller and state-specific trends attenuated the estimate. Negative-control cancer and cardiovascular mortality did not increase, motor-vehicle mortality moved in the opposite direction, and added injury/despair checks were mixed: fall mortality was positive but imprecise, while other non-transport injury and accidental poisoning did not increase. Sex/age stratification showed that the firearm-suicide association was concentrated among men. Total suicide also increased in the baseline model, while non-firearm suicide was positive but less stable. Firearm homicide showed no comparable association, although homicide-rate missingness was concentrated in small-population state-years. The evidence therefore supports a suicide-specific state-level association rather than a homicide, overdose, or uniform mortality-drift explanation; because political common support, pretrends, and state-trend sensitivity remain limiting, the defensible magnitude is smaller than the baseline 1.391 estimate.

Introduction

Permitless concealed carry is a recent policy margin inside a much older concealed-carry debate. Earlier right-to-carry studies focused mostly on transitions from no-issue or may-issue systems to shall-issue permitting, asking whether broader legal carrying deters violent crime or increases firearm violence [1–3]. That debate remains methodologically contested, and replication work has emphasized how sensitive concealed-carry estimates can be to event-study validity, heterogeneous treatment effects, and influential observations [4]. The policy landscape has also moved again. Since 2015, many states have gone beyond shall-issue permitting by allowing eligible adults to carry concealed handguns without first obtaining a carry permit, a shift that became even more salient after *Bruen* changed the constitutional environment for restrictive carry permitting [5]. The empirical problem is therefore no longer only whether permits should be granted more easily; it is what happens when the permit step itself is removed.

That distinction matters because the permit step can screen, delay, train, or document the person who wants to carry. A permitless-carry law may remove a training requirement, a permit-background-check requirement, a permit-specific misdemeanor screen, or a discretionary suitability review. It may also leave substantive firearm-possession prohibitions in place while eliminating the pre-carry application. Prior concealed-carry work has emphasized that these permitting provisions can matter for violence outcomes [2]. A recent permitless-carry working paper estimates increases in violent crime and firearm aggravated assault after adoption, reinforcing the need to evaluate the newer permitless margin separately from older shall-issue transitions [6]. Firearm-suicide work has also reported that concealed-carry permit requirements are associated with lower firearm suicide rates, alongside handgun purchase permits and waiting periods [7]. The policy margin is therefore plausible for suicide as well as homicide, but the relevant mechanism is not self-evident.

The suicide question deserves separate treatment from the homicide question. Firearm access is strongly associated with firearm suicide at household, individual, and state levels [8–10]. Purchaser-licensing evidence also links stronger handgun screening systems with lower firearm homicide and suicide in several state settings [11]. A law that changes public carrying may not directly change household gun ownership, but it can alter the carrying environment, the perceived normality of immediate firearm availability, and the screening requirements attached to carrying a handgun. Those pathways are closer to lethal-means access than to the deterrence theory that dominates older right-to-carry crime debates. For that reason, a permitless-carry mortality study should not treat homicide and suicide as interchangeable endpoints.

The main empirical risk is that permitless carry adoption is not random. States that adopted permitless carry in this period tended to have higher baseline gun ownership, higher rurality, higher Republican presidential vote share, and higher baseline firearm-suicide rates than states that did not adopt. These differences do not invalidate a panel design, but they do raise the standard for interpretation. A simple post-adoption coefficient can mix the policy association with pre-existing state trajectories, other firearm laws, political selection, health-access differences, substance-use environments, poverty, age structure, and mental-health workforce access.

Legal coding creates a second risk. Permitless open carry is not the same as permitless concealed carry, and several states do not fit cleanly into a binary adoption map. Vermont entered the panel already permitless, so it is not a within-panel adoption event. Arkansas has competing legal interpretations before and after its 2023 statutory clarification. Mississippi's 2015 and 2016 changes differ in how broadly permitless concealed carry applied. A paper that ignores these details can report a precise estimate attached to an imprecise treatment.

We address these two risks by combining a source-audited legal treatment file with a results hierarchy. The treatment audit records clean within-panel adopters, verified non-adopters, baseline-permitless states, partial cases, ambiguous reviewed cases, and policy-feature fields for training, carry-permit background checks, and permit-specific misdemeanor screening. The hierarchy specifies firearm suicide as the primary outcome, treats total suicide, non-firearm suicide, firearm homicide, and total firearm deaths as secondary outcome-family checks, and separates central estimation and sensitivity rows from exploratory mechanism rows. State-specific trends, cohort-window ATT summaries, event-time pretrend checks, Arkansas recoding, leave-one-adopter-out estimates, placebo timing, and covariate-layer sensitivity models are reported to define the boundary of the association rather than to multiply headline tests. Negative-control and despair-environment checks are part of that boundary: they test whether the firearm-suicide result is simply broad mortality drift, injury drift, or overdose-related mortality in adopter states.

This study treats permitless concealed carry as a distinct recent policy rather than as the endpoint of older right-to-carry classifications. It also uses a source-audited treatment file that separates clean within-panel adopters from baseline-permitless, partial, and legally ambiguous cases. Third, it evaluates suicide and homicide as separate mortality endpoints, with firearm suicide classified as the primary outcome and firearm homicide used as a secondary contrast. We ask whether states adopting permitless concealed carry experienced larger post-adoption changes in firearm suicide rates, whether that pattern is specific to suicide-related mortality rather than firearm homicide, and whether negative-control and accidental-poisoning outcomes support or undermine a broad mortality-drift explanation. Because adoption is politically and structurally patterned, the analysis emphasizes population weighting, state-specific trends, staggered-adoption summaries, event-time checks, Arkansas recoding, leave-one-adopter-out estimates, placebo timing, covariate-layer sensitivity models, and mortality-placebo checks as limits on the magnitude and interpretation of the association.

Data and Treatment Coding

State-year panel

The analysis uses an annual 50-state panel from 1999 through 2024. Mortality outcomes come from CDC WONDER underlying-cause mortality files and are expressed as crude rates per 100,000 residents [12]. Crude rates are the primary outcome scale because they measure the realized mortality burden in each state-year. Age-adjusted WONDER rates, standardized to the 2000 U.S. standard population, are used as demographic-composition sensitivity checks for firearm suicide, total suicide, firearm homicide, and total firearm deaths. Additional CDC WONDER exports provide cancer, cardiovascular, motor-vehicle, fall, other non-transport injury excluding falls and accidental poisoning, and accidental-poisoning mortality as negative-control or despair-environment checks, and firearm-suicide deaths by sex and ten-year age group for stratified checks. CDC WISQARS documentation provides the broader injury-surveillance context for fatal injury data derived from National Center for Health Statistics mortality files [13]. The primary clean-adoption models use 1,222 state-year observations from 26 source-verified within-panel adopters and 21 verified non-adopters. Firearm-homicide models use 1,142 observations because some state-year homicide rates are unavailable or suppressed. The primary outcome is firearm suicide. Secondary outcomes are total suicide, non-firearm suicide, firearm homicide, and total firearm deaths.

The baseline state-year covariates are unemployment and per-capita income. Sensitivity layers add external firearm-law indicators from the Tufts State Firearm Law Database [14], health-insurance access from Census SAHIE, drug-overdose mortality, Census demographic and poverty measures, and HRSA mental-health-provider access. RAND state-level household firearm-ownership estimates, rurality, presidential vote share, and baseline firearm-suicide rates are used for descriptive selection and heterogeneity analyses [15].

Treatment coding

The treatment is statewide permitless concealed carry: eligible adults can carry a concealed handgun in public without first obtaining a carry permit. The annual treatment indicator turns on in the first calendar year the statewide permitless concealed-carry rule is effective. Permitless open carry alone is not treated as permitless concealed carry.

The legal audit contains one row per state. It records 26 source-verified current-adopter rows, 21 verified non-adopter rows, one partial row, one ambiguous reviewed row, and one baseline-permitless row (Table 1). The primary analysis excludes the partial, ambiguous, and baseline-permitless rows so the headline estimate compares clean within-panel adopters with verified non-adopters. Mississippi, Arkansas, and Vermont remain visible in the legal audit rather than being hidden inside the binary treatment variable. Arkansas is tested separately under 2021 and 2023 coding, Mississippi is reintroduced in edge-case sensitivity runs, and Vermont is recorded as baseline permitless rather than as a within-panel adoption event. The full state-by-state treatment table in Supplementary Note 2 reports the state, adoption year, audit status, primary-map inclusion flag, coding note, and source URL trail.

Table 1: **Source-audited permitless-carry treatment map.**

Audit status	State count
Source-verified current adopter	26
Verified non-adopter	21
Ambiguous reviewed	1
Partial	1
Baseline permitless	1

The primary analytic panel contains 26 clean source-verified adoption-year states and 21 verified non-adopter states. Mississippi, Arkansas, and Vermont are handled outside the primary clean-adoption estimate.

The audit also records carry-permit features for the 26 source-verified adopter rows. Twenty-one had a training requirement removed, 25 removed a carry-permit background-check screen, and 12 removed a permit-specific misdemeanor-violence screen. These fields describe the carry-permit process, not every underlying firearm-possession prohibition. We therefore treat them as descriptive policy features rather than as definitive mechanism tests.

Empirical Strategy

The central specification is a two-way fixed-effects state-year regression:

$$Y_{st} = \alpha_s + \lambda_t + \beta D_{st} + X_{st}'\theta + \varepsilon_{st},$$

Here Y_{st} is the mortality rate for state s in year t, D_{st} equals one in and after the first calendar year statewide permitless concealed carry becomes effective, α_s and λ_t are state and year fixed effects, and X_{st} contains unemployment and per-capita income in the baseline model. Standard errors are clustered by state, with null-imposed Rademacher wild-cluster bootstrap p-values reported as a small-cluster inference check for the baseline TWFE rows. The coefficient β is the adjusted post-adoption association: the average within-state change after adoption, net of national year shocks and the baseline covariates. The unweighted model targets an average-state association; population-weighted models are reported separately because they answer the resident-weighted version of the same question.

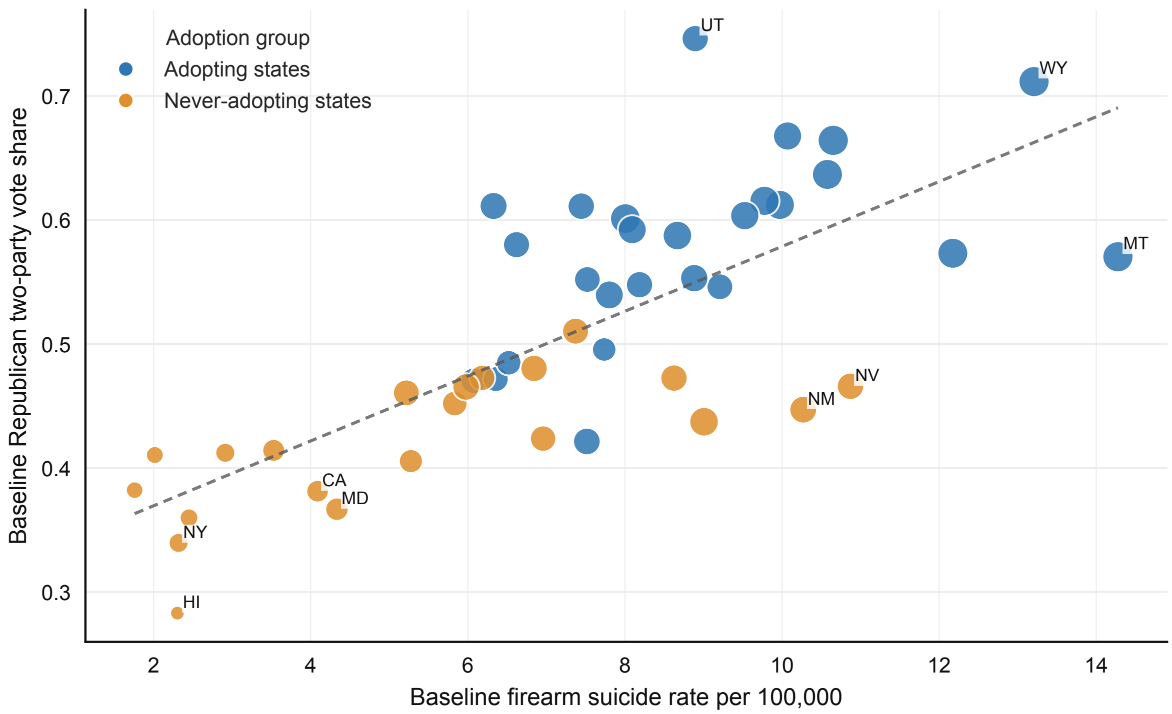
This estimand is familiar in firearm-policy panel studies, but staggered adoption complicates interpretation when effects vary across cohorts or time [16–19]. We therefore report cohort-window ATT summaries, not-yet-treated dynamic ATT estimates, stacked DiD estimates, and never-treated-control event-time fixed-effect estimates. We also estimate fractional-year treatment timing, which prorates the adoption year by the effective date of the law, and a covariate-balanced TWFE model that reweights non-adopter states toward adopter-state baseline firearm suicide, gun ownership, rurality, Republican vote share, income, and unemployment. These analyses ask whether the direction of the result is similar under alternative timing comparisons and whether pre-adoption event-time patterns undermine a parallel-trends interpretation [20].

The outcome ordering was fixed before drafting. Firearm suicide is primary. Total suicide, non-firearm suicide, firearm homicide, and total firearm deaths assess outcome specificity. Sensitivity analyses add firearm-law covariates, non-firearm contextual covariates, demographic and mental-health covariates, age-adjusted outcome rates, state-specific trends, COVID-period exclusions, Arkansas recoding, leave-one-adopter-out checks, and placebo timing among never-treated states. Cohort-window ATT estimates are reported as alternative staggered-adoption estimands, not as replacements for the baseline TWFE model. We do not treat every covariate layer as a superior causal model because some time-varying covariates may reflect broader state policy environments. Heterogeneity and policy-feature analyses are hypothesis-generating because subgroup and feature-specific adopter counts are small.

Results

Adoption Was Concentrated in Higher-Risk States

Before estimating mortality models, we compared adopter and non-adopter states on baseline state characteristics. Adopter states had higher Republican two-party presidential vote share (0.579 vs. 0.421), higher estimated household gun ownership (0.476 vs. 0.322), greater rurality (0.679 vs. 0.433), and higher baseline firearm-suicide rates (8.849 vs. 5.435 deaths per 100,000). The political baseline is the repository's covariate-balancing measure: the 2012 two-party Republican presidential vote share when available, or the latest available presidential vote-share value otherwise. Figure 1 shows that adopting states were concentrated toward higher baseline firearm-suicide rates and higher Republican two-party presidential vote share. This pattern shows that adoption was politically patterned and correlated with baseline firearm-suicide risk rather than distributed across comparable states.



Point size is proportional to baseline household firearm ownership. State-level baseline measures describe selection, not treatment effects.

Figure 1: **Selection into permitless carry adoption.** States that adopted permitless carry were concentrated toward higher baseline firearm-suicide rates and higher Republican two-party presidential vote share. Point size reflects estimated household gun ownership.

The Primary Association Was Concentrated in Firearm Suicide

Descriptive trends separated suicide from homicide before regression adjustment. Firearm-suicide and total-suicide rates were higher in adopter states, while firearm-homicide trends were noisier and did not visually track the same post-adoption pattern. These descriptive trends are not causal evidence because adopting and non-adopting states differed before adoption, but they motivate treating firearm suicide and firearm homicide as distinct endpoints.

The baseline fixed-effects model estimated a positive post-adoption association for firearm suicide (Table 2). Permitless carry adoption was associated with a 1.391 deaths per 100,000 increase in firearm-suicide rates (SE 0.222; $p < 0.001$). Total suicide also increased by 1.805 deaths per 100,000 (SE 0.294; $p < 0.001$), and total firearm deaths increased by 1.271 deaths per 100,000 (SE 0.482; $p = 0.008$). Non-firearm suicide was positive but smaller (0.415; SE 0.139; $p = 0.003$). Firearm homicide did not show a comparable association (-0.234; SE 0.338; $p = 0.489$). A null-imposed Rademacher wild-cluster bootstrap over the 47 state clusters produced similar inference for the baseline rows: $p = 0.001$ for firearm suicide, $p = 0.001$ for total suicide, $p = 0.009$ for non-firearm suicide, $p = 0.016$ for total firearm deaths, and $p = 0.509$ for firearm homicide. This baseline pattern shows that permitless carry adoption was associated most clearly with suicide-related mortality, not with firearm homicide. The result should not be treated as causal by itself because adopting and non-adopting states were not comparable before adoption.

Table 2: **Baseline fixed-effects mortality estimates.**

Outcome	Coef.	SE	p	N
Firearm suicide	1.391	0.222	< 0.001	1222
Non-firearm suicide	0.415	0.139	0.003	1222
Total suicide	1.805	0.294	< 0.001	1222
Firearm homicide	-0.234	0.338	0.489	1142
Total firearm deaths	1.271	0.482	0.008	1222

Because suicide mortality differs by age structure across states, we also estimated the baseline models using CDC WONDER age-adjusted rates (Table 3). The firearm-suicide estimate was nearly unchanged after age adjustment (1.407 vs. 1.391 deaths per 100,000). Total suicide and total firearm deaths remained positive, while firearm homicide remained statistically weak and below zero. This check supports the interpretation that the firearm-suicide result is not simply an artifact of different state age composition.

Table 3: **Age-adjusted outcome-rate sensitivity.**

Outcome	Coef.	SE	p	N
Firearm suicide, age-adjusted	1.407	0.228	< 0.001	1220
Total suicide, age-adjusted	1.914	0.327	< 0.001	1222
Total firearm deaths, age-adjusted	1.230	0.510	0.016	1222
Firearm homicide, age-adjusted	-0.230	0.426	0.590	1037

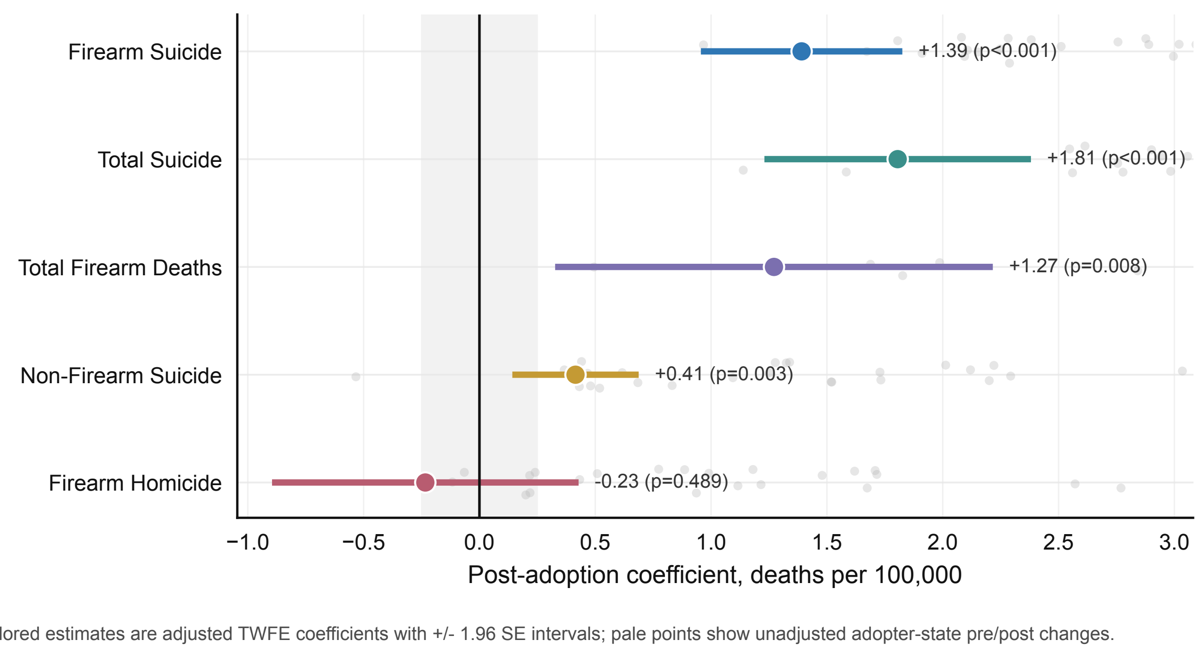


Figure 2: **Baseline fixed-effects coefficient forest.** Models include state and year fixed effects, unemployment, per-capita income, and state-clustered standard errors. The clearest baseline association is for firearm suicide and total suicide; firearm homicide is not positive.

The Firearm-Suicide Signal Persisted Across Diagnostics but Weakened Under Trend Tests

The firearm-suicide association was not limited to the baseline fixed-effects specification. In Table 9, the estimate remained positive when the model was population-weighted (0.919; $p < 0.001$), restricted to pre-COVID years (1.298; $p < 0.001$), estimated after excluding 2020–2021 (1.356; $p < 0.001$), and adjusted for external firearm-law covariates (1.096; $p < 0.001$). These checks change the estimand, time window, or measured policy environment, but they do not remove the positive firearm-suicide association.

A focused robustness table placed the baseline estimate beside checks that directly target timing, comparison-group, and outcome-specificity concerns (Table 4). Fractional-year treatment coding estimated a firearm-suicide association of 1.534 deaths per 100,000, indicating that the result was not created by treating mid-year laws as full-year exposure. The stacked DiD estimate was smaller at 0.665 deaths per 100,000, and the covariate-balanced TWFE estimate was 0.889 deaths per 100,000. A firearm-specific check that subtracts non-firearm suicide from firearm suicide retained a positive association (0.976; $p < 0.001$). These rows reduce the magnitude relative to the headline binary TWFE estimate, but they retain a positive firearm-suicide-specific association.

Table 4: **Primary firearm-suicide robustness estimates.**

Specification	Estimate	SE	p	N	Interpretation
Binary TWFE	1.391	0.222	< 0.001	1222	Headline annual treatment coding
Fractional-year TWFE	1.534	0.239	< 0.001	1222	Effective-year exposure prorated by law date
Stacked DiD	0.665	0.137	< 0.001	3446	Cohort stacks with not-yet-treated controls
Covariate-balanced TWFE	0.889	0.304	0.003	1222	Non-adopters reweighted toward adopter covariates
Firearm minus non-firearm suicide TWFE	0.976	0.226	< 0.001	1222	Firearm-specific association net of non-firearm suicide movement

Estimates are deaths per 100,000 residents. The first four rows estimate firearm suicide. The final TWFE row estimates firearm suicide minus non-firearm suicide, so it is a firearm-specific check rather than a total-suicide effect. TWFE rows include state and year fixed effects, unemployment, per-capita income, and state-clustered standard errors. The stacked DiD row uses 5-year pre/post cohort stacks with not-yet-treated controls.

Influence and placebo checks showed the same direction. Leave-one-adopter-out estimates ranged from 1.265 to 1.481, indicating that the baseline firearm-suicide estimate was not driven by a single adopter state. The observed firearm-suicide coefficient also exceeded the 95th percentile of the never-treated placebo-timing distribution, which supports the result as larger than the timing patterns generated by placebo adoption years among states that did not adopt. Legal edge-case checks retained the pattern: coding Arkansas as treated in 2021 or 2023 produced firearm-suicide estimates of 1.361 and 1.379, and reintroducing Mississippi alone produced an estimate of 1.300.

The main exception was trend sensitivity. With state-specific linear trends, the firearm-suicide estimate fell from 1.391 to 0.249 deaths per 100,000 and was not statistically significant at the 0.05 level ($p = 0.079$). Total firearm deaths also attenuated strongly under state trends (0.070; $p = 0.806$), while total suicide remained positive but smaller (0.372; $p = 0.025$). This attenuation means that part of the baseline association overlaps with longer state-specific firearm-suicide trajectories. The result therefore persists across several checks, but it is not robust to specifications that absorb separate linear trends for each state.

Additional covariate layers weakened the firearm-suicide estimate but did not change its direction. The estimate remained positive after adding firearm-law controls (1.096), health-access controls (0.707), overdose controls (0.492), and demographic-poverty controls (0.695). The mental-health-provider models are harder to interpret because they use only 141 state-years. In that short sample, the estimate was small and statistically weak, so those rows should be read as a sample-window limitation rather than evidence that mental-health-provider access explains the full-panel association.

Staggered-adoption estimates were also positive, but they were smaller than the baseline TWFE estimate. Cohort-window ATT estimates for firearm suicide were 0.601, 0.577, and 0.641 deaths per 100,000 across 2-, 3-, and 5-year windows. These estimates target different timing comparisons under heterogeneous treatment effects, so they should not be treated as simple replications of the TWFE coefficient. The direction of the association is consistent across TWFE and cohort-window ATT models, but the magnitude ranges from roughly 0.6 to 1.4 deaths per 100,000 depending on the estimand.

Table 5: **Staggered-adoption estimate summary.**

Outcome	ATT w2	ATT w3	ATT w5	Not-yet post	Interpretation
Firearm suicide	0.601	0.577	0.641	0.387	Positive, smaller than TWFE
Non-firearm suicide	0.270	0.239	0.241	0.077	Positive, smaller than TWFE
Total suicide	0.871	0.816	0.882	0.464	Positive, pretrend flag
Firearm homicide	-0.004	-0.107	-0.106	-0.157	No positive estimate
Total firearm deaths	0.382	0.435	0.493	0.200	Positive, pretrend flag

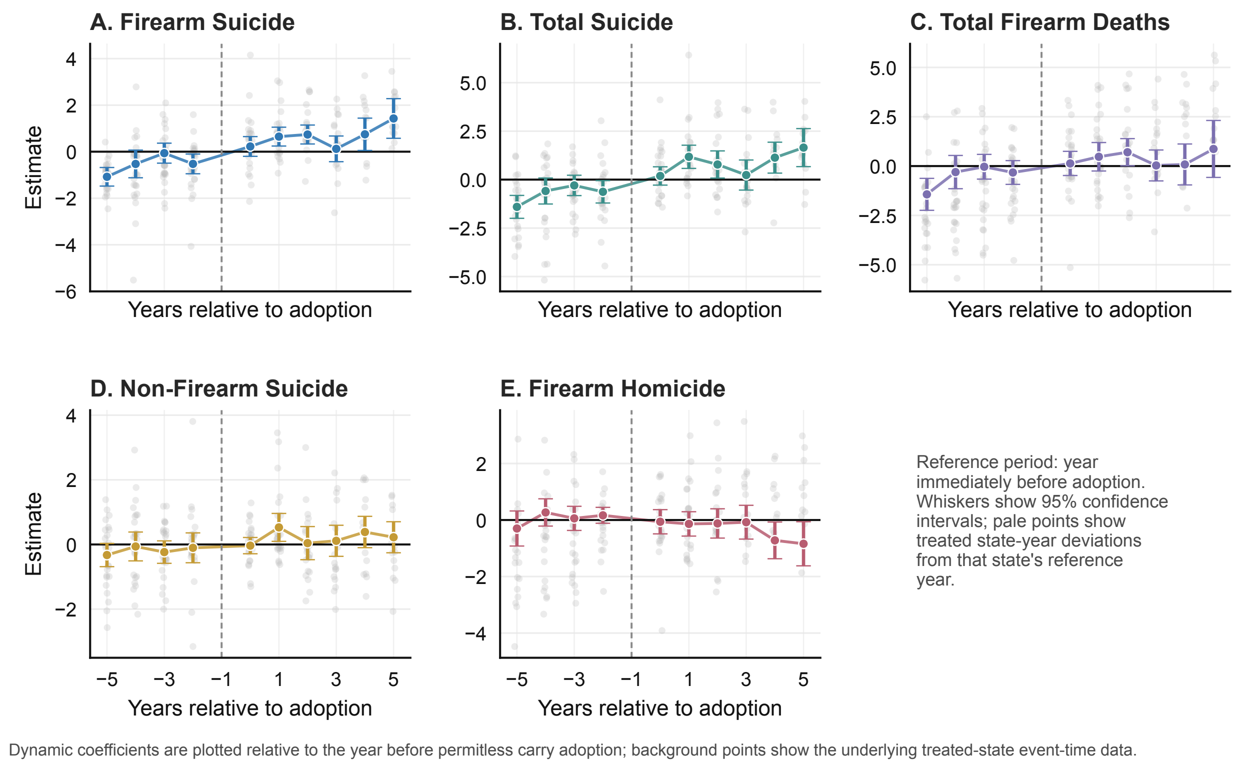


Figure 3: **Event-time diagnostic estimates.** Event-time models show post-adoption suicide-related increases but also pre-adoption signals for firearm suicide, total suicide, and total firearm deaths. Those pretrend flags prevent the staggered-adoption results from serving as confirmatory causal estimates.

Negative-control outcome models did not show a uniform positive mortality shift after permitless-carry adoption (Tables 6 and 7). Cancer mortality was positive but statistically weak (1.796 deaths per 100,000; $p = 0.441$), cardiovascular mortality was near zero (-0.591; $p = 0.919$), and motor-vehicle mortality moved in the opposite direction (-1.319; $p = 0.002$). The added injury and poisoning checks sharpen the interpretation. Fall mortality was positive but statistically weak (1.178 deaths per 100,000; $p = 0.197$), so the placebos are not perfectly inert. But after removing both falls and accidental poisoning from the non-transport bundle, other non-transport injury mortality was small and negative (-0.218; $p = 0.445$). Accidental poisoning mortality itself was negative and imprecise (-1.484; $p = 0.445$; 1,215 observations), which directly tests whether overdose-related mortality rose faster after adoption in adopter states. These rows do not prove exchangeability, but they make the main interpretation narrower: the firearm-suicide estimate is not well explained by homicide, overdose mortality, or uniform post-adoption mortality drift, while the positive fall row and trend sensitivity caution against treating the baseline 1.391 estimate as a clean causal magnitude.

Table 6: **Negative-control mortality estimates.**

Outcome	Coef.	SE	p	N
Cancer mortality	1.796	2.332	0.441	1222
Cardiovascular mortality	-0.591	5.810	0.919	1222
Motor-vehicle mortality	-1.319	0.416	0.002	1222

Table 7: **Additional injury and accidental-poisoning mortality estimates.**

Outcome	Coef.	SE	p	N
Fall mortality	1.178	0.913	0.197	1222
Other non-transport injury excluding falls/poisoning	-0.218	0.285	0.445	1222
Accidental poisoning mortality	-1.484	1.943	0.445	1215

Sex/age stratification made the firearm-suicide pattern more epidemiologically coherent (Table 8). The association was small and statistically weak among females (0.108; $p = 0.590$), but larger among males (1.665; $p < 0.001$). Male age-specific estimates were positive for ages 15–24, 25–44, 45–64, and 65+, with the strongest statistical evidence in the 25–64 adult groups. These rows are not mechanism proof because they remain state-level estimates, but they show that the primary association is concentrated in the demographic groups where firearm suicide is most common. Sex/age rows use complete broad-age cells after CDC WONDER suppression, so observation counts differ across strata.

Table 8: **Firearm-suicide estimates by sex and broad age group.**

Stratum	Coef.	SE	p	N
Female, modeled adult ages	0.108	0.200	0.590	749
Male, modeled adult ages	1.665	0.389	< 0.001	1141
Male, age 15-24	1.651	0.760	0.030	1085
Male, age 25-44	1.702	0.483	< 0.001	1038
Male, age 45-64	1.830	0.495	< 0.001	1066
Male, age 65+	1.332	0.737	0.071	631

Table 9: **Firearm-suicide specification and timing summary.**

Diagnostic row	Estimate	Diag. p	Interpretation
Baseline unweighted fixed effects	1.391	< 0.001	Clean-adoption state-level association
Population-weighted fixed effects	0.919	< 0.001	Resident-weighted association
Pre-COVID only	1.298	< 0.001	Drops post-2019 years
COVID-excluded	1.356	< 0.001	Drops 2020–2021
External firearm-law covariates	1.096	< 0.001	Policy-environment sensitivity
Leave-one-adopter-out	1.265–1.481	–	No sign changes across adopter exclusions
Never-treated placebo timing	$1.391 > 0.338$	–	Observed estimate exceeds placebo timing distribution
State-specific linear trends	0.249	0.079	Central attenuation caveat
Cohort ATT windows	0.601/0.577/0.641	–	Alternative staggered-adoption estimand
Not-yet-treated dynamic ATT	0.387	–	Alternative timing comparison
Event-time fixed effects	0.651	0.0000001	Positive post mean; diagnostic p is the minimum pre-adoption p

Estimates are deaths per 100,000 residents. TWFE rows include state and year fixed effects, unemployment, per-capita income, and state-clustered standard errors unless otherwise noted. Cohort-window and not-yet-treated ATT rows are alternative timing estimands. The event-time row is a pretrend diagnostic; its p column reports the minimum pre-adoption p rather than a post-adoption effect test.

Firearm Homicide Did Not Show the Same Pattern

Firearm homicide did not follow the suicide pattern. In Table 2, the firearm-homicide estimate was centered below zero (-0.234 deaths per 100,000; SE 0.338; $p = 0.489$). The age-adjusted homicide estimate was similar (-0.230; $p = 0.590$). Change-score windows also showed little evidence of a post-adoption increase: the adopter-minus-never differences were -0.034 deaths per 100,000 in the 2-year window ($p = 0.881$), 0.024 in the 3-year window ($p = 0.928$), and 0.035 in the 5-year window ($p = 0.899$). By contrast, firearm suicide was positive and statistically significant across the same 2-, 3-, and 5-year windows. In the staggered-adoption summaries, firearm homicide was not positive in either cohort-window or not-yet-treated comparisons.

Firearm-homicide missingness limits strong null claims. Of 1,222 possible clean-primary state-years, 80 homicide observations were missing or suppressed, and all missing observations were in the smallest-population quartile (Table 10). This missingness means the homicide result should be interpreted as no comparable evidence of change in the available state-year data, not as proof that permitless carry had no homicide effect.

Table 10: **Firearm-homicide missingness audit.**

Group	Possible	Observed	Missing	Missing pct.
All state-years	1222	1142	80	6.5%
Adopter-state state-years	676	610	66	9.8%
Non-adopter-state state-years	546	532	14	2.6%
Q1 smallest	312	232	80	25.6%
Q2	312	312	0	0.0%
Q3	286	286	0	0.0%
Q4 largest	312	312	0	0.0%

Missing firearm-homicide rates are concentrated in the smallest-population state-years. No state-year in population quartiles 2–4 is missing the firearm-homicide rate.

Taken together, the baseline and change-score results support the interpretation that the association after permitless carry adoption was more consistent for suicide-related mortality than for firearm homicide.

Feature Variation Was Too Limited for Mechanism Inference

The rurality results were the clearest subgroup pattern, but they do not explain the mechanism. The firearm-suicide association was larger in more rural states, but rurality is tied to many other state differences, including firearm ownership, politics, health access, and baseline suicide risk. For that reason, the rurality result shows where the association was larger, not why it occurred.

The data also do not observe the steps that would explain the association. We do not observe how often people carried firearms, whether they bought new firearms, how firearms were stored, whether a death followed an acute crisis, or whether the person who died had used permitless carry. Those missing links prevent a direct mechanism claim.

The policy-feature checks were also limited. Training removal, carry-permit background-check removal, and permit-specific misdemeanor screening are plausible pathways, but these features often change together. None of the feature-interaction estimates were statistically strong, and the background-check comparison could not be estimated because 25 of 26 verified adopters removed that screen. This is not only a data limitation; it reflects the structure of the current permitless-carry wave. These checks describe the laws; they do not show which part of the law explains the firearm-suicide association.

As a supplemental mechanism proxy, NICS handgun background-check data do not provide clean evidence of a net acquisition increase. The direct handgun-check proxy increased after adoption in the 1999–2021 NICS panel (476.952 checks per 100,000; $p = 0.010$), but permit checks moved in the opposite direction and the combined handgun-or-permit proxy was not statistically distinguishable from zero (-1797.590; $p = 0.338$). Excluding 2020–2021 gave the same qualitative pattern: the handgun row remained positive, while the combined handgun-or-permit row remained statistically weak (-976.463; $p = 0.409$). NICS checks are not firearm sales, and the locally parsed official FBI file did not provide clean 2022–2024 state-year rows. We therefore interpret the NICS result as consistent with permit-to-point-of-sale reclassification under permitless carry, not as evidence that adoption increased firearm acquisition.

Discussion

This study evaluates a specific, recent form of concealed-carry deregulation: the removal of the carry-permit requirement. It does not estimate the full history of right-to-carry laws, and it does not test whether public carrying deters crime. In this 50-state panel, the clearest pattern was outcome-specific and bounded: permitless carry adoption was more consistently associated with suicide-related mortality than with firearm homicide, accidental poisoning, or a uniform rise across negative-control mortality outcomes.

The firearm-suicide estimate was positive in the baseline fixed-effects model, age-adjusted outcome models, change-score windows, population-weighted models, Arkansas recodings, leave-one-adopter-out checks, and several covariate and staggered-adoption estimates. Total suicide often moved in the same direction, while total firearm deaths were less useful as a headline outcome because the row also carried pretrend and trend/covariate-balance fragility: the baseline total-firearm estimate was 1.271 deaths per 100,000 ($p = 0.008$; wild-cluster $p = 0.016$), but it fell to 0.070 under state-specific trends ($p = 0.806$) and 0.141 under covariate balancing ($p = 0.851$). Firearm homicide, other non-transport injury, and accidental poisoning did not show comparable positive associations. This contrast is the central empirical result: the signal is most consistent on firearm suicide, not on violence overall, overdose-related mortality, or mortality generally.

The baseline TWFE and cohort-window ATT estimates differ in magnitude. Under heterogeneous treatment effects, the cohort-window ATT estimates avoid some of the weighting problems that can affect TWFE estimates, while the TWFE model remains the baseline adjusted state-year association. Both approaches produce positive firearm-suicide estimates, but the range is wide: approximately 0.6 deaths per 100,000 in cohort-window ATT models and 1.391 deaths per 100,000 in the baseline TWFE model. State-specific trends reduce the estimate further. We therefore interpret the direction and outcome specificity of the association as more stable than the baseline magnitude.

The positive non-firearm suicide association introduces additional ambiguity. A pure firearm-method substitution account would predict firearm suicide rising while non-firearm suicide remains stable or falls. Instead, both moved upward in the baseline model, although the non-firearm result was smaller and less stable across sensitivity layers. This weakens mechanism specificity without reversing the primary association. The pattern is consistent with broader suicide-risk differences in adopter states, policy effects operating through channels beyond immediate firearm access, or noise in a secondary outcome.

The main limitation is identification. State-specific linear trends attenuated the firearm-suicide estimate, and event-time models showed pre-adoption signals for firearm suicide, total suicide, and total firearm deaths. Negative-control cancer, cardiovascular, fall, other non-transport injury, and accidental-poisoning outcomes do not show a uniform positive mortality shift, but they are noisy rather than perfectly inert: cancer and fall mortality are positive and statistically weak, while motor-vehicle, other non-transport injury, and accidental poisoning move in the opposite direction. These patterns rule out a simple generalized-mortality explanation, but they also show why the paper should not claim a precise causal effect. Adopting states were already on different mortality trajectories, and policy timing may have coincided with other changes not fully captured by the covariates. The estimates should therefore be read as associational evidence for a firearm-suicide-specific pattern, not proof that permitless carry caused a 1.391 deaths-per-100,000 increase.

Covariate balancing reduced baseline differences in firearm suicide, gun ownership, rurality, income, and unemployment, but political overlap remained incomplete. Using the 2012 political baseline described above, a common-support audit found that 21 of 26 adopters had Republican two-party vote shares above the maximum observed among verified non-adopters. The same audit is baseline-definition-dependent: using a 1999–2014 mean gives 20 of 26 adopters above the non-adopter range. Either definition leaves roughly four-fifths of adopters outside the non-adopter political support range. The balanced estimate therefore improves the comparison but does not create a fully exchangeable set of adopting and non-adopting states.

The legal audit is an important part of the study because treatment coding is itself a source of uncertainty. Arkansas, Mississippi, and Vermont illustrate the problem. A state can be baseline permitless, partially permitless, ambiguously interpreted, or newly clarified by statute. A state can also remove the pre-carry permit application while retaining underlying possession prohibitions. The audit makes those distinctions visible and prevents the treatment variable from appearing more certain than the law allows.

The feature analysis is best read as a negative design result. Training removal, background-check removal, and misdemeanor-screen removal are plausible pathways, and prior concealed-carry work has emphasized the importance of permit provisions [2]. In this dataset, however, feature groups are small or nearly universal among clean adopters. The absence of statistically significant feature interactions is therefore not strong evidence against those pathways; it shows that the current adoption wave does not provide enough cross-state variation to test them well.

These findings fit a broader firearm-policy literature that increasingly treats suicide and homicide as distinct outcomes. State firearm laws associated with lower homicide are not always the same as those associated with lower suicide [21, 22]. Recent suicide-specific evidence links handgun permits, waiting periods, and concealed-carry permit requirements to lower firearm suicide rates [7]. The present analysis extends that logic to the recent permitless-carry wave and finds that the clearest mortality association is again on the suicide side.

Several limitations remain beyond identification. The panel is state-level and cannot identify individual behavior, carrying frequency, firearm acquisition, storage, mental-health crises, crisis timing, or interpersonal circumstances. Some homicide observations are missing because of suppression or incomplete availability, with missingness concentrated in small-population state-years. The HRSA mental-health-provider specifications use a short recent window, and the policy-feature fields require deeper statute-level auditing before they can support mechanism claims.

The practical implication is primarily methodological. Permitless-carry studies should separate primary outcomes from secondary and exploratory checks, audit legal treatment timing, report edge-case coding, report both unweighted and population-weighted estimands, distinguish staggered-adoption estimands from event-time pretrend diagnostics, and test whether the result is duplicated in overdose, injury, homicide, or broad mortality controls. Under that standard, the strongest statement from this analysis is cautious: adopting states experienced larger post-adoption increases in firearm suicide rates, the pattern was not reproduced by homicide, accidental poisoning, or uniform mortality controls, and the defensible magnitude is bounded by trend sensitivity and incomplete political common support.

Additional Methods Details

All baseline fixed-effects models used ordinary least squares with state and year fixed effects, unemployment, per-capita income, and state-clustered standard errors. Robustness specifications excluded 2020–2021, restricted the panel to pre-2020 years, applied population weights, added state-specific linear trends, added external firearm-law covariates, prorated first-year treatment exposure by the law's effective date, or reweighted non-adopter states toward adopter-state baseline covariates. Covariate-layer sensitivity models added firearm-law domains, uninsured rates, overdose mortality, age structure, race and ethnicity shares, poverty rates, and mental-health-provider access as described in the repository output tables. Negative-control models used the same baseline TWFE specification for cancer, cardiovascular, motor-vehicle, fall, other non-transport injury excluding falls and accidental poisoning, and accidental-poisoning mortality. Sex/age models collapsed CDC WONDER ten-year age groups into 15–24, 25–44, 45–64, and 65+ strata and used complete broad-age cells after suppression. The unweighted estimates target the average state-level association; the population-weighted estimates target the average resident-level association.

Change-score analyses computed, for each adopter state, the difference between the average post-adoption rate and the average pre-adoption rate in 2-, 3-, and 5-year windows. Never-adopter states were assigned comparable window summaries and compared with Welch tests. Staggered-adoption analyses included cohort-window ATT estimates, not-yet-treated dynamic ATT estimates, stacked DiD estimates, and never-treated-control event-time fixed-effect estimates. Age-adjusted outcome models used CDC WONDER age-adjusted rates as alternate outcomes and retained the same state and year fixed effects, unemployment, per-capita income, and state-clustered standard errors as the crude-rate baseline model. These checks were designed as sensitivity analyses because several outcomes showed pre-adoption event-time signals.

The outcome order was fixed before drafting the paper. Firearm suicide is the primary outcome. Total suicide, non-firearm suicide, firearm homicide, and total firearm deaths are secondary outcome-family checks. Cancer, cardiovascular, motor-vehicle, fall, other non-transport injury excluding falls and accidental poisoning, and accidental-poisoning mortality are negative-control or despair-environment diagnostics. Firearm-law covariates, contextual covariate layers, age-adjusted outcome rates, staggered-adoption estimators, Arkansas recoding, leave-one-out influence, and placebo timing are sensitivity analyses. Rurality, gun ownership, baseline-risk heterogeneity, sex/age stratification, and policy-feature interactions are exploratory.

Data and Code Availability

Processed analysis outputs, source-audited policy coding files, figure-generation scripts, and the manuscript-ready result tables are maintained in the project repository. Raw CDC mortality files and firearm-law source files are governed by their originating agencies and are documented in the repository source appendix.

Author Contributions

B.K. and Y.W. assembled the policy audit, implemented the mortality panel analyses, generated figures and tables, interpreted the results, and drafted the manuscript.

Competing Interests

The authors declare no competing interests.

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